

Tacit: the AI Apprentice

An apprentice that learns a desk job by watching an expert do it, asks why at the right moments, maps it into a Work Map, and teaches the next person. Hack-Nation × ElevenLabs, 7th Global AI Hackathon, Challenge 01.

THE PROBLEM

Experienced people retire with decades of judgment that was never written down. Screen recordings show *what* was clicked, not *why*; the limits, exceptions and moments to stop and ask live only in people's heads. New hires learn them by breaking them.

WHAT WE BUILT

- **Capture.** The expert shares their screen in a floating pill. A vision model turns frames into events ("cost center 4711 → 0400"). An ElevenLabs voice agent stays quiet while they type, read or talk, and asks short questions at natural pauses. At least three, one about a guardrail, checked in code.
- **Map.** A spoken debrief closes the gaps, then a short teach-back the expert confirms. The result is a clickable Work Map: every step and guardrail linked to its screen moment, a clip, and the expert's own words.
- **Teach.** A voice tutor watches a new hire work a case the expert never showed, locks the practice ERP's save before a wrong decision is posted, and explains it with the expert's reasoning. It ends with mastered / practice next.
- **Sia on the Work Map.** A guide agent walks a learner (or reviews with the expert) step by step, focusing nodes as it talks.

BEYOND THE BRIEF

- **Two experts, one task:** compare two sessions, see what differs, and ask each expert why in their next session.
- **Any language:** 72 languages; the Work Map stays in English with the original quotes.
- **Agent-ready export** with every guardrail as a stop rule, and an **autopilot** that posts routine invoices and hands every judgment call to a person.
- **Living Work Map:** a repeat session asks only about what changed. **Coverage check:** notices a case no map covers.

THE APPRENTICE TEST

When to ask	A floor in the pill: voice activity, pixel motion and reading time hold the floor; only a pause after an action lets the agent speak. Pace follows Settings, never below three.
What to ask	Each pause carries the latest screen events; a fast model skips what the screen or common practice already answers ($\geq 90\%$ sure).
When it has understood	The draft map's open questions drive the debrief; the teach-back is the proof, and only the expert's "yes" saves the map.
Whether the new hire learned	A case the expert never showed; the report is computed from what the learner did, not by an LLM.
Trust	Off the record (O); an on-device privacy shield paints over personal data before frames leave the machine; names redacted with Presidio.

HOW IT'S BUILT

Desktop app (Electron, React, TanStack, @xyflow) for macOS, Windows and Linux. **Backend** (FastAPI, Python) on Render behind an access key, with Postgres and Storage on Supabase. **Voice:** ElevenLabs Agents (apprentice, tutor, guide) with Scribe realtime and client tools; vision and merging with OpenAI. **Quality:** 482 backend and 262 UI tests, run on every change.

DATA

No external datasets. Demos run on **Ledgerly**, a built-in practice ERP with synthetic supplier invoices (no real people or companies).

MOONSHOT

A living company memory: every expert and workflow in one map that stays current, asking only about what's new, so people keep the judgment calls and agents take the routine steps safely.